

Hosi Reanspiration Pro User Guide : Below is a detailed guide to optimizing your workflow in REAPER.

1. Preparation & General Interface

1. Copy the entire Hosi_Reanspiration_Pro.lua file and the reanspiration_*.lua files into the same folder in REAPER/Scripts/Hosi Reanspiration (or similar).
2. In REAPER: Actions → Show action list → ReaScript → Load... select Hosi_Reanspiration_Pro.lua , then assign a keyboard shortcut if desired.
3. Run the script, and the “Reanspiration – by Hosi” window will open with the following tabs: Generation, Performance, Creative Tools, Drums, Arranger, Settings.

Some common points:

- At the beginning of each session, select **the Language** (English / Vietnamese / ...) and **Theme** in the header if needed.
- Always **select a MIDI item** before pressing the Generate/Apply buttons; otherwise, the script will report an error: "No MIDI item selected...".
- **Each function** has its own Undo system (Undo Humanize, Undo Melody, Undo Bass...), so feel free to try them out.

2. Tab Generation – creating chords and background bass

2.1 Setting the scale and progression

1. Select an empty MIDI item (or one with chords if you want to analyze it again).
2. In the **Generation tab** :
 - Root Note (the letter re, mi, etc.) : select the root note, or leave it as "Random" if you want it to be random.

- Scale Type : select Major, Natural, Minor, Dorian...; there is a **Detect Scale button** to analyze the selected MIDI word.
- Progression :
 - "Random" → enter Number of Chords so the tool can automatically generate the chord progression.
 - Alternatively, choose a preset (Pop, Jazz, Blues, etc.) from the Library.
- Complexity : 0 = triad, 1 = 7th, 2 = 9th, ... 5 = altered.
- Add Secondary Dominants : Turn this on so the tool has a 50% chance of inserting the V7 of the next chord.

2.2 Bass pattern & modulation

- Bass Pattern : Choose the standard bass pattern (Root, Root-5, Alberti, Funk, Walking, etc.).
- **Modulation** section :
 - Tick Enable Modulation , select Target Root , Target Scale , Modulate at Chord n , and Use Pivot Chord if you want smooth key changes using common chords.

2.3 Seed – Fixed Random

- Seed section :
 - Lock button: lock/unlock current seed (keep the result).
 - The "lightning bolt" button: creates a new seed and locks it immediately.
 - Number field: Enter seed manually if you want to reproduce the exact results of the previous session.

2.4 Creating and managing chords

- Click Generate Chords & Bass to create chords and bass for the selected item (use channel 1 for chords, channel 2 for bass by default).

- **Add Note** : Randomly inserts a new note based on the scale being analyzed (useful for testing).
- **Delete Chords – Keep Bass/Melody** : removes the chord notes but keeps the bass and melody.

2.5 Chord Viewer

- **Chord Viewer** section :
 - **Analyze & Show Chords** : analyzes the chords and scale of the item.
 - **Clear** : Clears the analytics cache.
 - There is a popup displaying a list of chords and scale names.

3. Performance tab – adjust voicing & humanize

1. **Transpose** : Change all items by ± 2 octave using a combo, apply immediately to MIDI.
2. **Spread** : 0 = close, 2 = widen the distance between notes in the chord.
3. **Voicing** : None / Drop2 / Drop3 / Drop4 / Drop2&4; reapply voice-leading to the existing chord.

Humanize section :

- **Groove Template** : None/Random/8th Swing/16th Swing.
- **Swing %** : sometimes select groove swing.
- **Velocity Accent** : None / Strong 1&3 / Weak 2&4 / Syncopated.
- **Timing - PPQ , Velocity - %** : the strength of humanize.
- **Humanize Notes** button to apply, and the **Undo Humanize** button to revert.

4. Creative Tools tab – melody, ostinato, advanced bass, arp/strum, expression, rhythm

4.1 Melody Generation & Motif

1. In the **Melody Generation** section :

- **Contour** : Choose the melodic form (Random, Ascending, Arch, Descending, Valley, Markov Chain, Counterpoint, Thematic Motif).
 - If you select **Markov Chain** , click **Learn from Selection** to have it learn the style from a sample melody item.
 - **Melody Pattern** : None (algorithm), or select a pattern from the Library (tension/release, etc.).
 - **Density** : 1–10; the higher the density, the more notes.
 - **Min/Max Octave** : distance limit.
2. Click **Generate Melody** → the script analyzes the chord track or item, then writes the melody to the Melody channel (default 3).
 3. **Generate Motif** : Creates a one-measure motif based on the first chord, reusable when Pattern = None + Contour “Thematic Motif”.
 4. **Generate Counter-Melody** : Creates a counter-melody based on the current melody in the Counter-melody channel.
 5. There are buttons for **Undo Melody** and **Undo Counter-Melody** .

4.2 **Melody Mutator**

- **Invert Pitch** , **Retrograde** , **Displace Rhythm (PPQ)** , and **Undo Mutation** button .
- **Workflow**: Create melody → Mutator to quickly generate variations (suitable for fills and variations).

4.3 **Cinematic Ostinato**

1. Select pattern length (1/4 bar, 1/2 bar, 1 bar) and **Build-up %** (velocity increase over time).
2. Click **Generate Ostinato** : the script uses the current chords and scale to create an ostinato cinematic on the Melody channel.

3. Undo Ostinato to return to the previous stack.

4.4 Bass (creative) & Lock to Kick

1. In **Bass Generation** : the Bass Pattern combo lists all patterns plus an option called "Walking Bass Algorithmic".
2. Lock to Kick Drum : if enabled, the script searches for the track “Drum” and the Kick notes (36), creating a bass line according to **the rhythm of the kick** (root note according to the chord).
3. Press Generate Bass , use Undo if needed.

4.5 Arpeggiator & Strummer

1. Mode : None / Arpeggiator / Strummer.
2. If Arp:
 - Arp Pattern (Up, Down, UpDown, Random), Arp Rate (1/4, 1/8, 1/16, 1/32, Random).
 - Euclidean options: Steps, Hits.
3. If Strum:
 - Strum Pattern (from library), Strum Delay (PPQ) , Up-strum Velocity % , Palm Mute , Guitar Voicing Open , Up-Stroke Latency Scale .
4. Click Apply Arp/Strum → apply it to the selected notes (or all notes if nothing is selected). There is an Undo Arp/Strum option .

4.6 Expression Engine

- In Dynamic Expression Engine :
 - Dynamics % : modulation width CC1/CC11.
 - Follow Pitch Swell , Breath Phrase End Dip .
 - Legato Overlap (ms) + Smart Sustain Pedal .
- Click Apply Expression , Undo if necessary.

4.7 Rhythm Applicator

- **Rhythm Pattern** : Choose from the Library (Ballad, Pop Syncopation, EDM Sidechain, Bossa, Waltz, etc.).
- **Apply Rhythm** : Apply this pattern to the chords (recreate the rhythmic division). It has its own Undo function.

5. Tab Drums – pattern + Euclidean sequencer

1. **Drum Pattern** : choose preset (Basic 4/4, Techno, Hip-hop, Reggae, etc).
2. **Press Generate Drums** : to create a beat for the entire length of the item on channel 10.
3. **Generate Fill** : creates a fill in the last measure; **Create Variation** : adds/removes random kicks/hats.

Euclidean Sequence :

- **Instrument** : Choose from Drum Map (Kick, Snare, HH, Tom...).
- **Steps , Hits , Rotation** : Set up the Euclidean pattern.
- **Click Generate Layer** : this will create a Euclidean layer for that instrument, deleting the old notes of the corresponding pitch before inserting them.

6. Arranger tab – Arranger + Song Architect + Arranger King Bridge

6.1 Song Architect (create your own structure & full song)

1. In the **Arranger tab** , under **Song Architect Structure** :
 - **Style Template** : Choose a template (Pop, EDM, Cinematic, etc.) from the list. Each template has blocks: Intro, Verse, Chorus, etc., each with its own energy.

- Select:
 - Clear Existing Regions : Delete the old region before creating the new one.
 - Clear Existing Tracks : (optional) deletes existing tracks and creates new ones from the template.
 - Use Existing Chords : If enabled, it will use your existing chord tracks instead of generating new ones.
 - Auto-Mixing Volume : Creates automated volume based on the energy of each section.
- 2. Create Structure Regions : Create regions using the template, and update the region table below.
- 3. Generate Full Song :
 - Create tracks for chords, melody, bass, and drums (from a .RTrackTemplate track template if available), and generate MIDI for each region based on the energy profile.
 - Apply automation volume according to energy.

Timeline preview (energy graph):

- You can drag nodes on the graph to adjust the energy for each section; Song Architect uses this energy when generating.

6.2 Arranger – Regions & Pattern Pool Management

- Sync from Project Regions : reads all existing regions and populates the table.
- Display panel includes: ID, Name, Energy, Start, Length, Actions.
- In the Actions column:
 - If the region is already "glued," the word "Glued" will appear; if not, there is a "Glue " button for each region or "Glue All Regions ."

- **Pattern Pool** : You can save items as patterns , rename them , delete them , add them , move them up/down , and remove them to manage and reuse patterns.
- **Output Track Name** , **Create Regions for Sections** , **GENERATE ARRANGEMENT** : create a new arrangement from the pattern pool.

6.3 Arranger King Bridge

- **Section “Arranger King Bridge”**:
 - Select the “Guide Track” item from Arranger King, then click Scan Guide Track .
 - The script reads text events / analyzes note pitch clusters to deduce the structure (verse/chorus/bridge/...); creates corresponding regions in the project.

7. Settings tab – routing, chord source, Drum Map, seed, Custom Menu

1. MIDI Channel Assignments :

- Chords , Bass , Melody ; counter-melody always = Melody + 1; Drums fixed on channel 10.

2. Chord Source :

- Tick " Read chords from a specific track ," and enter the track name (default "Chords"). From here, all creative tools (melody, bass, rhythm, etc.) will take the harmonic context from this track instead of the selected item.

3. Drum Map Editor :

- Drum Map for Kick/Snare/Tom/HH...; re-enter the note number that matches your VSTi.

4. Seed System (repeated from the Generation tab, but also available in Settings language strings).

5. Custom Menu :

- The “Custom Menu...” button opens a popup for configuring your own menu: add Actions, Sub-menus, Separators; drag and drop to reorder; save to ExtState.

8. Smart Presets & Preset Browser

- The header has a **Smart Presets button** : open **the Smart Preset Browser** .
- Left column: Factory Presets are grouped by genre (Pop, EDM, Hip-hop, Jazz, Cinematic, Game, Experimental...). Selecting a preset will apply a set of settings (scale type, complexity, groove, drum pattern, bass pattern, voicing, etc.).
- Right column: User Presets – you save a new preset using the “Save Current Vibe” button, enter a name and genre.
- Right-click user preset to delete.

Here's a quick workflow suggestion for you.

1. **Generation** tab : select scale / progression / complexity → Generate Chords & Bass.
2. **Drums** tab : select Drum Pattern → Generate Drums.
3. Tab **Creative Tools** :
 - Melody Generation (Contour + Markov or Motif).
 - Apply Rhythm / Arp / Strum → Humanize → Expression.
4. If you want the full song: go to **Arranger** → **Song Architect**, select a template → Create Structure → Generate Full Song.

Here's the specific "recipe": from a blank project → a 100 BPM, 16-bar long Pop song using Reanspiration Pro in REAPER.

1. Prepare the project and script.

1. Open REAPER, create a new project, set the tempo to 100 BPM, and the time signature to 4/4.
 2. Make sure you have installed and loaded Hosi_Reanspiration_Pro.lua (Action List → Script).
 3. Run the script, and select Vietnamese in the header if desired.
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2. Create a background MIDI item for the 16-bar section.

1. Create a new track and give it the temporary name "Song Sketch".
 2. Drag the time scale to measure 17 (bar 17) and then Insert → New MIDI item to create MIDI items from bar 1–17 (16 bars).
 3. Select this item, open Reanspiration Pro (if it's closed).
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3. Generate basic Pop chords and bass

3.1 Setting the Scale and Progression

Generation tab :

1. Root Note: Choose C (easy to listen to, pop).
2. Scale Type: select “Major”.
3. Progression:
 - Select a Pop preset such as “Pop 1 I–V–vi–IV” or “Modern Pop I–V–vi–iii” from the progression list.
4. Number of Chords:

- If you use a preset to fix the length of each chord, let it set itself (this cell will be disabled).
 - For a simple 16-bar Pop song, you can use a 4-chord progression, each chord lasting 1 bar, looped 4 times (the script will be distributed across the entire piece).
5. Complexity: Choose 1 (7th) for a slightly "richer" pop, or 0 for a triad.
 6. Uncheck "Add chords from the decade" if you want a clean look, or turn it on if you prefer a bit of tension.

3.2 Bass pattern & seed

1. Bass pattern: select "Pop - Pushing 8ths", "Pop - Synth Octaves" or "Pop Bassline I-VII-vi-IV" from the list.
2. In the Seed section: if you want the result to repeat, click the padlock to **lock the current seed** ; if you want a random result, click the create new seed button first.

3.3 Generate

1. Make sure MIDI item 16 bar is still selected.
2. Click Generate Chords & Bass.
3. Verify this by opening the MIDI editor:
 - The default Chords Channel is 1.
 - Bass on channel 2.

4. Create a Pop 4/4 drum and fill it.

the Drums tab :

1. Still selecting MIDI item 16 bar on the track "Song Sketch".

2. Drum Pattern: Choose the preset "Pop - Four on the Floor" or "Standard 4/4" (from the Library).
 3. Click Generate Drums. The script will:
 - Create a drum beat on channel 10 within that item.
 4. Click Generate Fill to manually add a fill to the last measure (bar 16).
 5. If you want a slight swing/variation, use Create Variation to randomly add/remove Kick/Hi-hat.
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5. Create a Pop melody for 16 bars.

Switch to the **Creative Tools tab** , then **to the Melody Generation section** .

1. Make sure the script is reading the chords from the newly created **Chords** /item track: in the Settings tab, leave the Chord Source as default (read from the selected item).
2. Contour: Choose "Arch" or "Ascending" for a fresh Pop; if you like, you can try "Markov Chain" later.
3. Melody Pattern: Leave it as "None" (the algorithm will automatically generate tension/release) or choose a pop melody pattern from the Library.
4. Density: Choose 4–6 for just the right number of notes.
5. Lowest/Highest octave:
 - Min Octave 4, Max Octave 5 (vocal/broad synth region).
6. Click Generate Melody. The script will:
 - Analyze the chords/scales in the item.

- Write the melody on **the Melody channel** (default 3) spanning bars 1–16 along the selected contour.
7. If you want to add color:
- Press Generate Counter-Melody to create a secondary melody (channel 4).
 - If you don't like the melody, click Undo Melody and then Generate again with a new seed.
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6. Apply rhythm, arp/strum & humanize

6.1 Rhythm for chords

1. Still in the **Creative Tools tab** , **Rhythm Applicator section** .
2. Rhythm Pattern: select “Pop - Modern Syncopation” or “Pop - Piano Ballad”.
3. Click Apply Rhythm:
 - The script removes sustain chords and reconstructs them according to the rhythmic pattern, preserving the harmonic content.

6.2 Arp or Strum (depending on the instrument used)

If you plan to layer piano/keyboard:

1. In **Arpeggiator Strummer** :
 - Mode: select “Arpeggiator”.
 - Arp Pattern: “UpDown” or “Random”.
 - Arp Rate: 1/8 or 1/16.
2. Click Apply Arp/Strum.

If you want to layer guitar:

1. Mode: Strummer.
2. Strum Pattern: choose the preset type “Folk D-DU-DU” or “Pop Syncopation DU...”.
3. Adjust Strum Delay (PPQ) to ~20–40, turn on Guitar Voicing and Palm Mute slightly if needed.
4. Apply Arp/Strum.

6.3 Humanize & Groove

Go to **the Performance tab** :

1. Groove Template: Choose “8th Swing” for a slight groovy effect, or “None” for a straighter look.
2. Velocity Accent: “Strong 1&3” or “Syncopated” depending on feel.
3. Timing - PPQ: 5–10; Velocity - %: 10–15.
4. Click Humanize Notes; if it's too strong, Undo Humanize and then reduce the parameters.

7. Expressive notes for track lead/pad

Creative Tools tab , Dynamic Expression Engine section :

1. In MIDI Editor, select the track/item you intend to use for **the lead** or **pad** (for example, copy the melody to the "Lead" track).
2. Dynamics: approximately 60–80.
3. Turn on Follow Pitch Swell and Breath Phrase End Dip if it's a wind/string instrument; it works quite well with a synth pad too.
4. Legato Overlap: 20–40 ms; Smart Sustain Pedal is on if it's a piano.

5. Click Apply Expression.

8. Use Song Architect for 16-bar structures (advanced option)

If you want to separate a 16-bar song into distinct Intro/Verse/Chorus sections and have a multi-track feel:

1. Create Regions:

- Region 1: Bars 1–5 → “Intro”.
- Region 2: Bars 5–9 → “Verse”.
- Region 3: Bar 9–13 → “Chorus”.
- Region 4: Bar 13 –17 → “Verse/Outro”.

2. **Arranger** tab , then to **the Song Architect Structure section** :

- Style Template: select Pop template.
- Tick "Use Existing Chords" to use the harmonies you generated on the "Song Sketch" track.
- Uncheck Clear Existing Tracks if you don't want it to delete the current track; or enable it if you want the script to automatically create the track arrangement “Chords/Melody/Bass/Drums” based on the template.
- Auto-Mixing Volume: Enable this to automate volume generation based on the energy consumption of each region.

3. Click Generate Full Song:

- The script creates tracks with chords, melody, bass, drums, and distributes MIDI by region with different energy levels (light intro, stronger chorus, etc.).

If you only want to keep it a simple, basic 16-bar section, you can skip the Song Architect step.

9. Track arrangement & rendering

1. Separate the parts:
 - Copy MIDI chords to the track “Piano Chords”, route to VSTi piano.
 - Melody to “Lead Synth / Vocal MIDI”, different VSTi route.
 - Bass retains or copies to the “Bass” route synth/bass guitar.
 - The drums are sent to the VSTi drum/sampler, with notes mapped using Drum Map if needed.
 2. Add FX, sidechains, and mixes like in a familiar workflow.
 3. Render the 16-bar track as a stereo mixdown.
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